

SHAP · LIME · Attention

WHAT'S INSIDE

- 01** Interpretability vs explainability
- 02** SHAP — Shapley values for ML
- 03** LIME — local linear approximations
- 04** Attention visualisation
- 05** Natural-language final step

"Credit score" isn't an answer. SHAP gives one — "CIBIL 640 × debt-to-income 0.72 × late payments last 12mo drove the decline." That is explainable AI. Every high-stakes model in India needs it in 2026.

When was the last time a system gave YOU an actual reason — not just a code?

WHY THIS LESSON MATTERS

HDFC declines your loan. Why?

NUMBER 1

640 ×

From real-world deployments.

NUMBER 2

0.72 ×

From real-world deployments.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Interpretability vs explainability.

2

Apply SHAP to tabular models.

3

Apply LIME to text or tabular.

4

Visualise attention in transformers.

5

Natural-language explanations.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

Interpretability vs...

Two different questions

02

**SHAP — Shapley
values for ML**

Game theory applied to features

03

LIME — local linear...

Model-agnostic

04

**Attention
visualisation**

Where the transformer looks

01

PART 1 OF 4

Interpretability vs...

Two different questions

Interpretability vs explainability

Interpretability: is the model itself understandable?

Explainability: can we explain a complex model's decision?

Linear regression is interpretable; DL is (post-hoc) explainable.

Regulators want explainability; debuggers want interpretability.

BOTH MATTER

Different needs

Know which you need.

DEFINITION

Interpretability vs explainability

Two different questions

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

SHAP — Shapley values for ML

`shap.TreeExplainer(model)` for gradient boosted models.

`shap_values = explainer.shap_values(X_test)`.

Global: summary plot.

Local: force plot for a single prediction.

Mathematically principled; handles interactions.

GOLD STANDARD

SHAP

Most battle-tested post-hoc method.

KEY POINTS

- `shap.TreeExplainer(model)` for gradient boosted models
- `shap_values = explainer.shap_values(X_test)`
- Global: summary plot
- Local: force plot for a single prediction

LIME — local linear approximations

Perturb input around a point.
Query black-box on perturbations.
Train weighted linear model.
Coefficients = local explanation.
FAST
Any model
Trade globalness for flexibility.

DEFINITION

LIME — local linear approximations

Model-agnostic

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Attention visualisation

Extract attention matrices layer-by-layer.

Visualise with BertViz, attention rollouts.

Useful for NER debugging, LLM factuality.

CAVEAT: attention $\neq$ explanation — model may "look" somewhere and not use it.

WATCH

Attention $\neq$ proof

Supplementary signal only.

KEY POINTS

- Extract attention matrices layer-by-layer
- Visualise with BertViz, attention rollouts
- Useful for NER debugging, LLM factuality
- CAVEAT: attention $\neq$ explanation — model may "look" somewhere and not use it

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Take ONE prediction; write 3 sentences

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which model of yours most needs explanations?

Q2

Would you use SHAP or LIME for tabular GBM?

Q3

Where does attention mislead you?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

SHAP is based on:

A

Bayesian

B

Shapley values

C

Backprop

D

Random forest

SHAP is based on:



CORRECT ANSWER

Shapley values

WHY

Game-theoretic Shapley values

LIME is:

A Global

B Local

C Both

D Neither

LIME is:



CORRECT ANSWER

Local

WHY

Local linear approximation per point

Interpretable model example:

A

DL

B

Linear regression

C

Random forest

D

Transformer

Interpretable model example:



CORRECT ANSWER

Linear regression

WHY

Coefficients directly tell the story

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Interpretability: is the model itself understandable?
Explainability: can we explain a complex model's decision?
Linear regression is interpretable; DL is (post-hoc) explainable.
Regulators want explainability; debuggers want interpretability.
BOTH MATTER
Different needs
Know which you need.

THE TAKEAWAY

"If you can't explain a decision to a 10-year-old, you can't explain it to a regulator."

WHAT TO NOTICE

- 1 Interpretability vs...
- 2 SHAP — Shapley values for ML
- 3 LIME — local linear...
- 4 Attention visualisation

SHAP for W3 model

Real-estate predictions.

DELIVERABLES

- TreeExplainer.
- Summary plot.
- 3 predictions explained.
- 1-page PDF.

WHAT 'GOOD' LOOKS LIKE

Deliverable: PDF on forum.

ONE THING TO REMEMBER

“

"If you can't explain a decision to a 10-year-old, you can't explain it to a regulator."

— AI Learning Hub

END OF LESSON 8

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 4 — AI Security.